Case Study: Building Simulated Clients for a Multi-Turn Agent Eval

AI accountant benchmark | User simulation, measurement checks and grading design

Note: client data, case details, people and internal figures are anonymized or omitted. Examples are invented.

1. Summary

As a Research Intern at Keeper, I worked on a benchmark in which an AI accountant interviews a simulated client and prepares a complete federal and state tax return. I designed the simulated clients, co-designed the grading, and checked whether the scores measured what they claimed.

- **Persona playbook and build pipeline** used to write dozens of client personas from real, anonymized tax records.
- **Rubric logic co-designed;** automated grading reached **96%+ agreement** with hand marks.
- **Grading cost cut ~60%** by changing what the grader reads, not which model judges.
- **Cross-model error analysis** that separated real model differences from grading and logging artifacts.

2. The Core Problem: The Client Controls the Information

In a multi-turn eval the simulated user is not scenery. It decides what the agent can learn, and only if the agent asks. A client that is too helpful does the agent's work: when three models hit one client's refund exactly, the transcripts showed the client reciting its own totals. A client that is too vague makes the test unfair. The persona's job is to make every failure *fair*.

> **Priority order:** fairness first, realism second, variety last. A rule enters the playbook only if it makes the test truer or fairer.

3. The Method: From Tax Records to a Persona Memo

Step	What happens
1. Trace	Tie every figure on the answer-key return back to the client's documents. Sort each fact: volunteered, known only if asked, or only in the files.
2. Write	A memo in the client's own voice: life facts, documents, worries, and short "has-to-ask" lines for facts that surface only when the accountant asks the right question.
3. Check	Every fact sourced, nothing invented to hit a number; no circularity (the memo must not leak the answer); a careful accountant can still reach the right return.
4. Run & read	Run several models and read transcripts, not just scores, before a memo counts as done.

4. Making Failures Fair, and Different

- **Mechanics vs. behaviors.** A mechanic applies to every client (e.g., send the bank export rather than recite totals). A behavior is a per-client setting with no default.
- **Personality exposes different failures.** A quiet client punishes models that don't ask; partial answers punish bundled questions; a confidently wrong client punishes deference. If every client is terse, the bench tests one skill.
- **Planted noise is the test.** Duplicate rows and plausible wrong answers stay in, as long as the right answer stays reachable.

5. Checking the Measurement

Check	What it caught
Perfect scores = warning	Exact refunds from three models traced to a client leaking totals, not to model skill.
Blind review only	One question to a reviewer ("did you read my case notes?") moved a score from 84 to 70. Every review since runs in a fresh agent with fixed inputs.

Cheap experiments first	A $0.19 test replaced a planned feature; a test scoped from 16 runs to 4 still found a wording problem across the memo set.
Audit the leaderboard	In one grading pass, a single zero score and one unreadable PDF decided the model ranking.
Measure the hypothesis	An apparent "answer dumping" behavior turned out to be a logging artifact.

6. Grading Design

- **Atomic rubrics:** independent, weighted yes/no decisions rather than a tree where one answer gates the next; roughly 35 questions at most.
- **Graded, not gated:** zeroing an unfileable return hides variance that is useful for training. Continuous scores, with large penalties for grave errors.
- **The sanity check:** if a weak return outscores a thoughtful one, the rubric is broken. Fix that, not the decimals.
- **Four outcomes per hidden fact:** found and used, found but treated wrong, guessed (no credit even when right), missed.
- **Cost:** grading a text version of the return with caching and batching cut cost ~60% without changing the judge.

7. What Carries to Any Domain

User simulation for agent evaluation applies wherever an AI must interview someone: medical, legal or financial intake, tutoring, recruiting. The simulator moves scores more than the judge does, so measure it, give it a fidelity check, keep its personalities varied, and read transcripts before trusting a number.